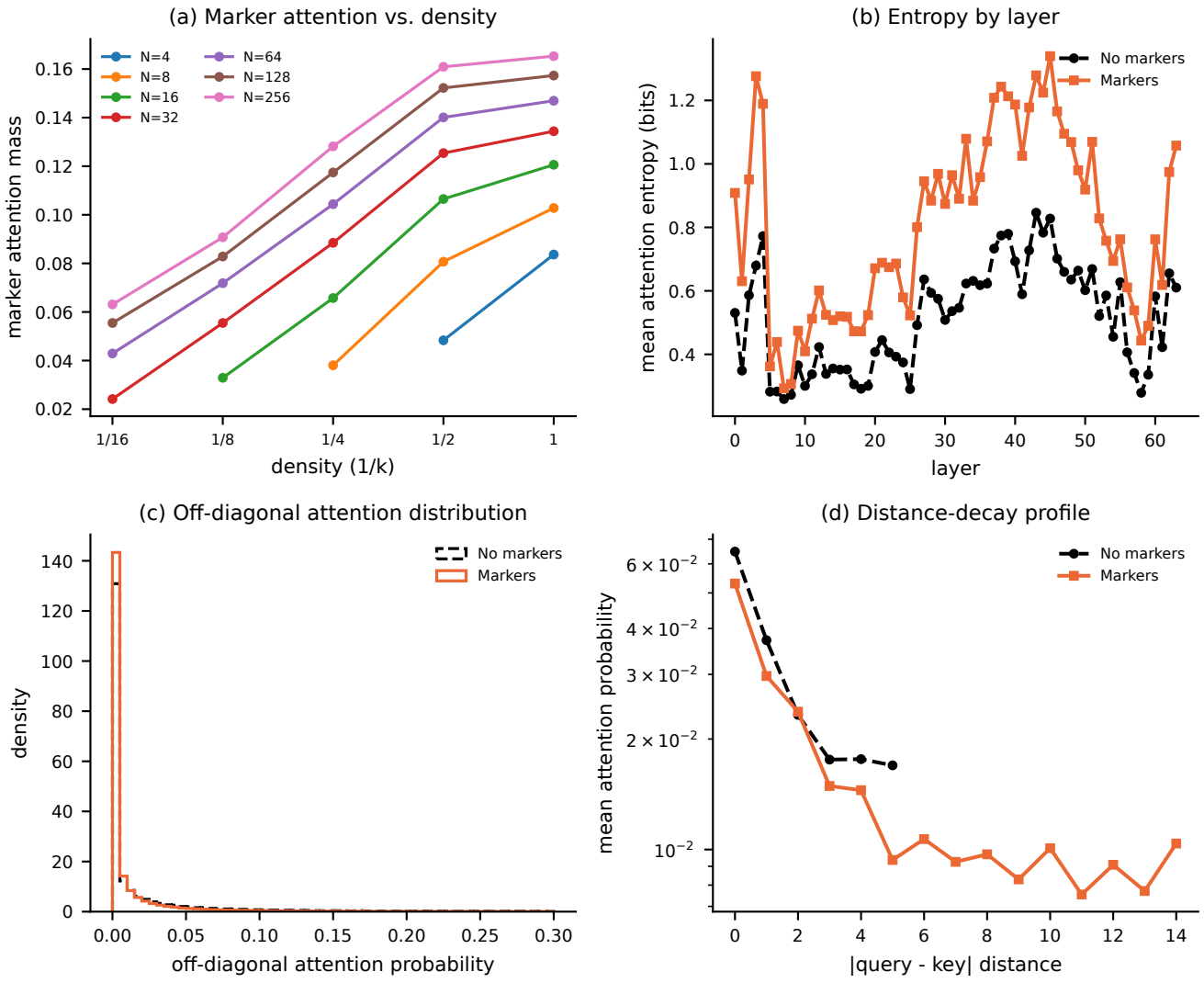

(a) Marker attention vs. density
N=4
N=8
N=16
N=32
N=64
N=128
N=256
marker attention mass
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
1/16
1/8
1/4
1/2
1
density (1/k)
(b) Entropy by layer
No markers
Markers
mean attention entropy (bits)
1.2
1.0
0.8
0.6
0.4
0
10
20
30
40
50
60
layer
(c) Off-diagonal attention distribution
No markers
Markers
density
140
120
100
80
60
40
20
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
off-diagonal attention probability
(d) Distance-decay profile
No markers
Markers
mean attention probability
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
0
2
4
6
8
10
12
14
|query - key| distance